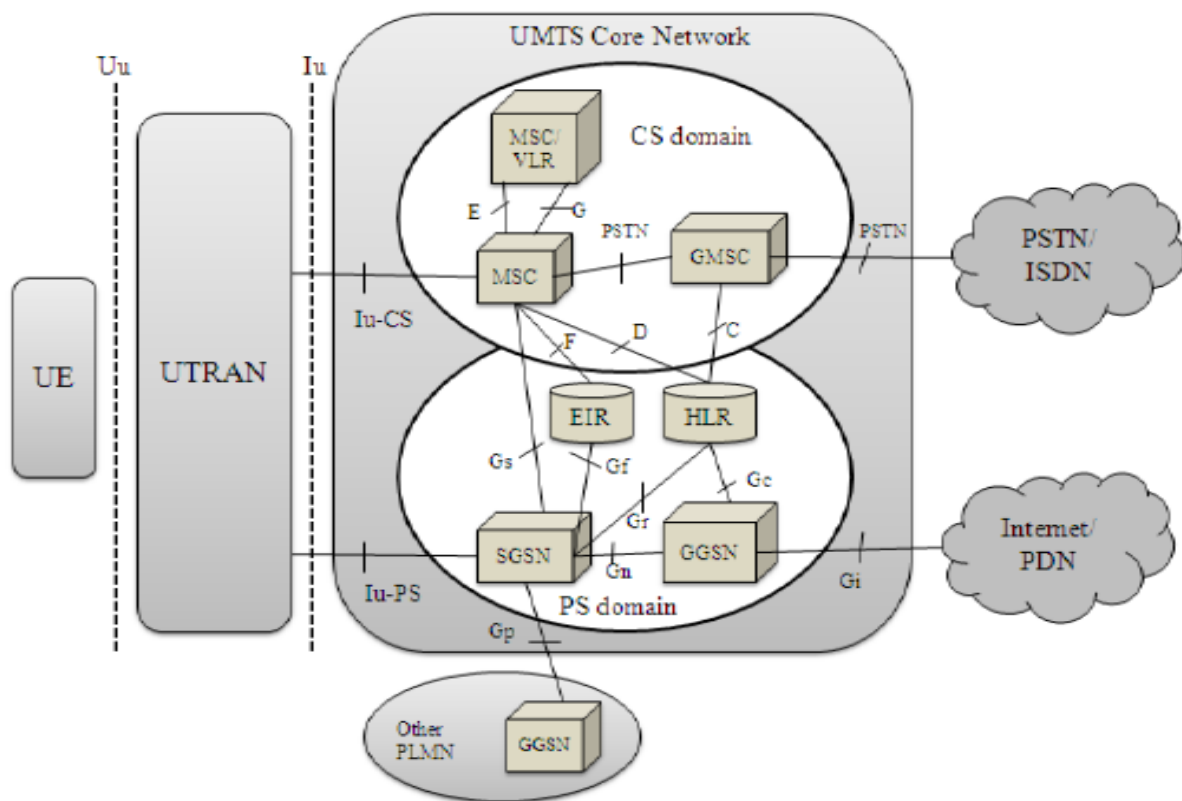


Universal Mobile Telecommunication System

The Universal Mobile Telecommunication System (UMTS) is one of the most popular IMT-2000 standards. The UMTS represents an advancement compared to 2.5G technology in terms of data services and speed. The UMTS network can broadly be divided into User Equipment (UE) domain, UMTS Terrestrial Radio Access Network (UTRAN) domain, and core network domain.

Architecture of UMTS Core Network

The core network of UMTS network is responsible to control the Packet Switched (PS) and Circuit Switched (CS) service when the user initiates or terminates the call. In addition, the core network also provisions the external networks' internetworking, mobility of the user within the home network and to the visited foreign networks that includes location and handoff management, charging and accounting, and authentication of the wireless client.



Serving GPRS Support Node

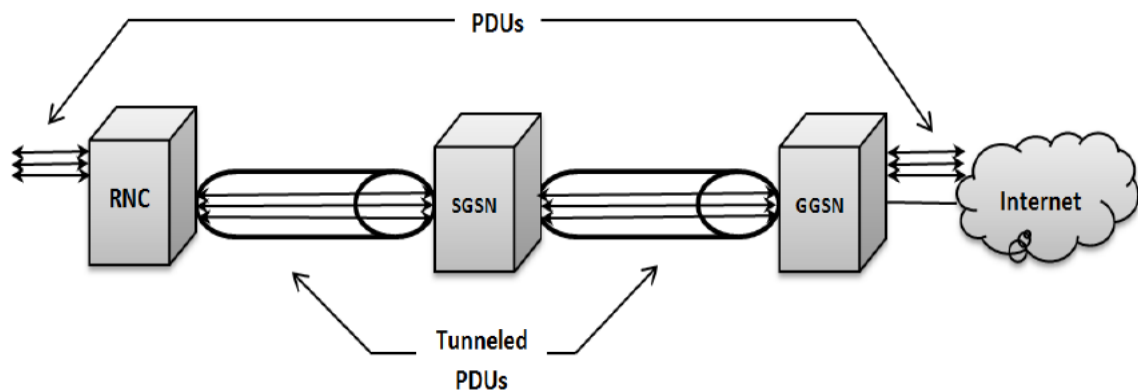
The Serving GPRS Support Node (SGSN) is a gateway within the UMTS network that performs various services including data routing within its service region. The functionalities of SGSN encompass:

- ☐ Authentication
- ☐ QoS management
- ☐ Charging function and
- ☐ Location tracking within UMTS network

Gateway GPRS Support Node:

The Gateway GPRS Support Node (GGSN) is the last component in the UMTS network. The external PDNs and the UMTS backbone network (IP network and radio network) are internetworked via

GGSN. In order to initiate a data session with the external networks, the UE must obtain the IP address which is used to recognize it by the external networks. For this, the UE has to establish a PDP context activation. The GGSN is responsible to assign the IP address to the mobile user; moreover, it serves as a default gateway to the UMTS UEs. Once the PDP context is activated, the UE sends the data packets to the RNC first which encapsulates the data packets and send them to the GGSN via SGSN. The GGSN decapsulates the data packets and forward them to the PDN. In the same way, when the packets are coming from the external network, first they are encapsulated by the GGSN and forwarded to the RNC via SGSN. The RNC decapsulate the data packets and send them to the UE. It is worth recalling that the GGSN assigns the IP addresses to the UEs. Therefore, the UE's IP address has the identical network prefix as the GGSN has.



UMTS Terrestrial Radio Access Network Domain

The radio resource management is performed by the UMTS Terrestrial Access Network (UTRAN). As illustrated in Figure 2.11, the UTRAN is composed of two network entities i.e., Radio Network Controller (RNC) and Node-B.

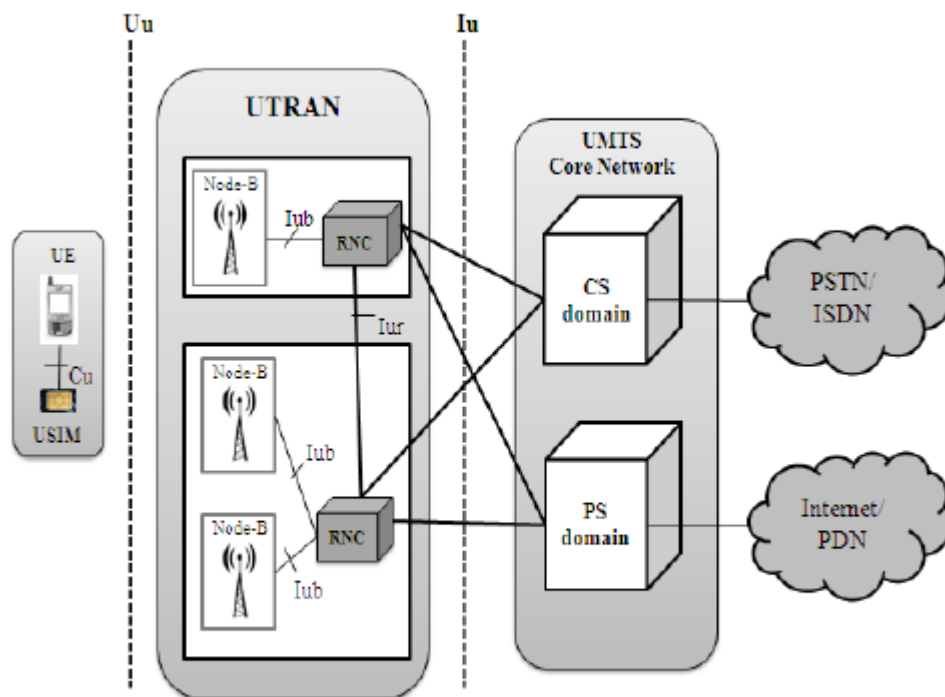


Figure 2.11: UMTS UTRAN architecture

Radio Network Controller

The Radio Network Controller is connected with the Iu-PS and Iu-CS interfaces with the UMTS PS and CS domains, respectively. Within UTRAN, it is connected with the Node-B with the Iu-b interface. The main functionality of the RNC is to control the Node-Bs and handle the data transmission over the UMTS radio interface. This includes radio resource management, intra RNC handoff, and encryption of data to/from the UEs etc.

Node B

The Node-B is the hardware that provides the radio connection to communicate directly to the UE. It contains transmitters and receivers to communicate to the mobile users. In the cellular communication, the UEs located within the same or different cells are not allowed to communicate directly, however, they need Node-B to pass on the information.

UMTS Data Session Establishment Mechanism

For the acquirement of UMTS packet switched services, the UE must be authenticated and known (in terms of its routing and location information) to the UMTS network. The presence of UE is informed to the SGSN by using the GPRS attach procedure. Moreover, in order to establish the PS data connection with the UMTS core network so that the data can be sent and received by the UE, Packet Data Protocol (PDP) context is activated. By using the PDP context, the UE is known by the GGSN and the external networks. The following subsections represent the GMM attach and PDP context activation mechanisms along with the RAB assignment mechanism.

GMM Attach Procedure

The GMM attach procedure is a three way hand-shake between the UE and SGSN. Without the GMM attach procedure, UE cannot be reachable by the network because the SGSN does not have the routing and location information of UE. This procedure makes the UE known to the network and SGSN collects the authentication, location management and access control parameters input from the UE.

The GMM attach procedure is used for the PS signaling connection between the UE and the SGSN are illustrated in the Figure 2.12. The GMM attach procedure is initiated when the UE sends (1) *GMM: Attach Request* to the SGSN. The *GMM: Attach Request* contains the location information, IMEI, and International Mobile Subscriber Identity (IMSI). Upon receipt of the *GMM: Attach Request*, the SGSN performs the authentication procedure by forwarding the sent information to the HLR. When the HLR verifies the given entries, the SGSN replies the *GMM: Attach Request* with the (2) *GMM: Attach Accept* message to the UE. The accept message indicates that the UE is allowed to access the UMTS network resources. Finally, the UE sends the (3) *GMM: Attach Complete* message which completes the GMM attach procedure.

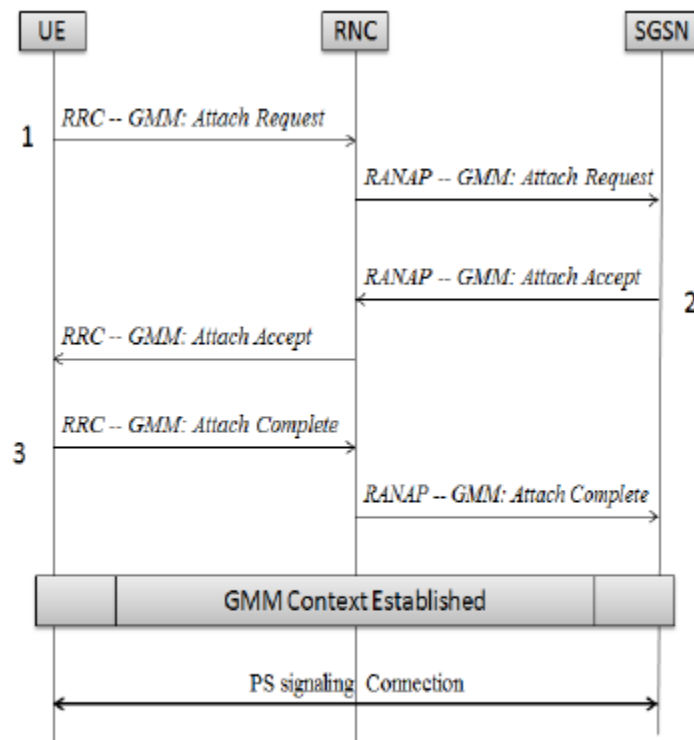


Figure 2.12: GMM attach procedure

PDP Context Activation Procedure

In order to initiate the data session with the external Packet Data Networks (PDNs), the UE has to perform the PDP context procedure. Actually, after establishing a GMM attach procedure, the UE is only known to and authenticated by the SGSN node. However, to communicate with the external packet networks, UE has to initiate the PDP context. This internetworking with the external PDN is arranged by the GGSN. The GGSN keeps the routing information of PS-attached UE. The GGSN creates the tunnel to send the data to the UE to its current point of attachment.

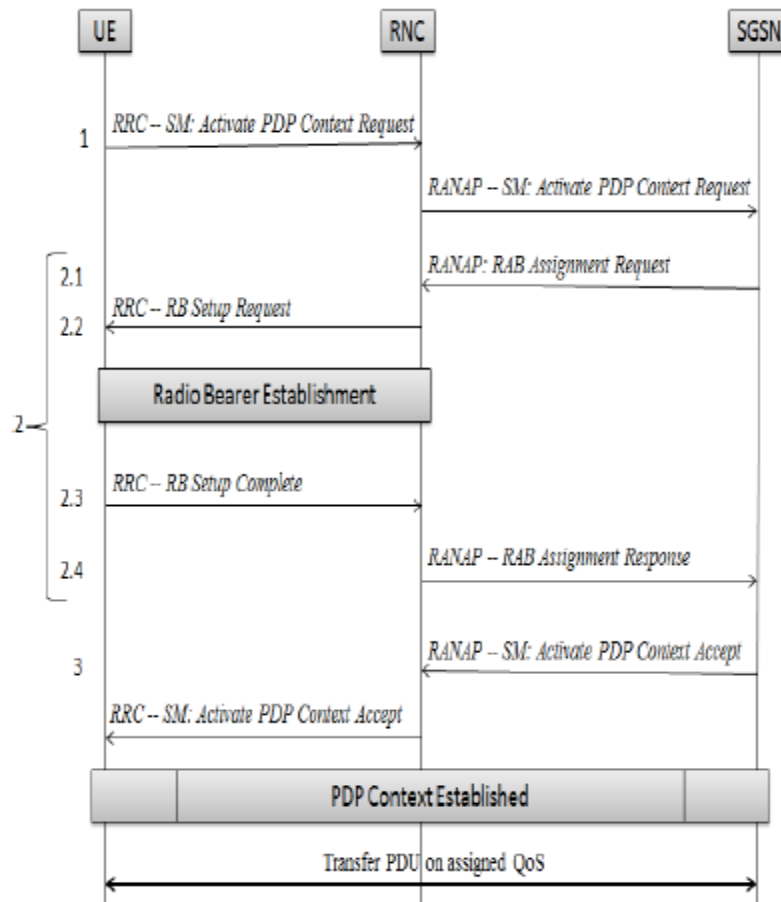


Figure 2.13: PDP context activation procedure

Figure 2.13 demonstrates the PDP activation mechanism. When the UE wants to send the data and still it does not have the PDP context activated, it sends (1) *SM: Activate PDP Context Request* to the SGSN via RNC. When the SGSN receives this message, it initiates the (2) RAB assignment process, which is discussed in the next subsection. Upon receipt of (2.4) *RAB Assignment Response*, the SGSN requests the GGSN to establish a PDP context on the basis of negotiated parameters. The GGSN allocates an IP address for the UE and sends it to the SGSN. Upon receipt, the SGSN forwards the GGSN assigned IP address to the UE by using the (3) *SM: Activate PDP Context Accept* message.

RAB Assignment Procedure

The step 2 in the Figure 2.13 illustrates the Radio Access Bearers (RABs) assignment procedure. When SGSN receives (1) *SM: Activate PDP Context Request* then it initiates the RAB assignment process. During this step, the SGSN investigates that whether the network has the enough resources to facilitate the required QoS request. On receipt of (2.1) *RANAP: RAB Assignment Request*, the RNC inspects the availability of the downlink and uplink capacity. If the RNC can facilitate the UE with the requested QoS, then it sends a (2.2) *RRC: RB Setup Request* to the UE to establish an appropriate Radio Bearer (RB). If the requested QoS is not available at the RNC, then the RNC sends the *RRC: RB Setup Request* with the negotiated QoS parameters. The UE establishes the assigned RB as requested by the RNC and send back a (2.3) *RRC: RB Setup Complete*. Finally, (2.4) *RANAP: RAB Assignment Response* is sent to the SGSN as a RAB-RB mapping notification.